PROBLEM 48

Two Connected Particles, One on Paraboloid

Original statement · Problems.pdf, p. 33

Multi-Particle Manifold

Two heavy particles P and Q, each of mass m, are connected with an inextensible, massless string of length ℓ. Particle P is constrained to stay on a paraboloid, given implicitly by equation $z = cR^2$, where c is a known constant and we have introduced the usual cylindrical coordinates $\{R, \theta, z\}$. The string passes through a small hole at the origin O, and particle Q is restricted to hang along the vertical axis. The reference frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial. There is no friction in this problem. Assume that the string is long enough so that Q never reaches O and that it is always tense.

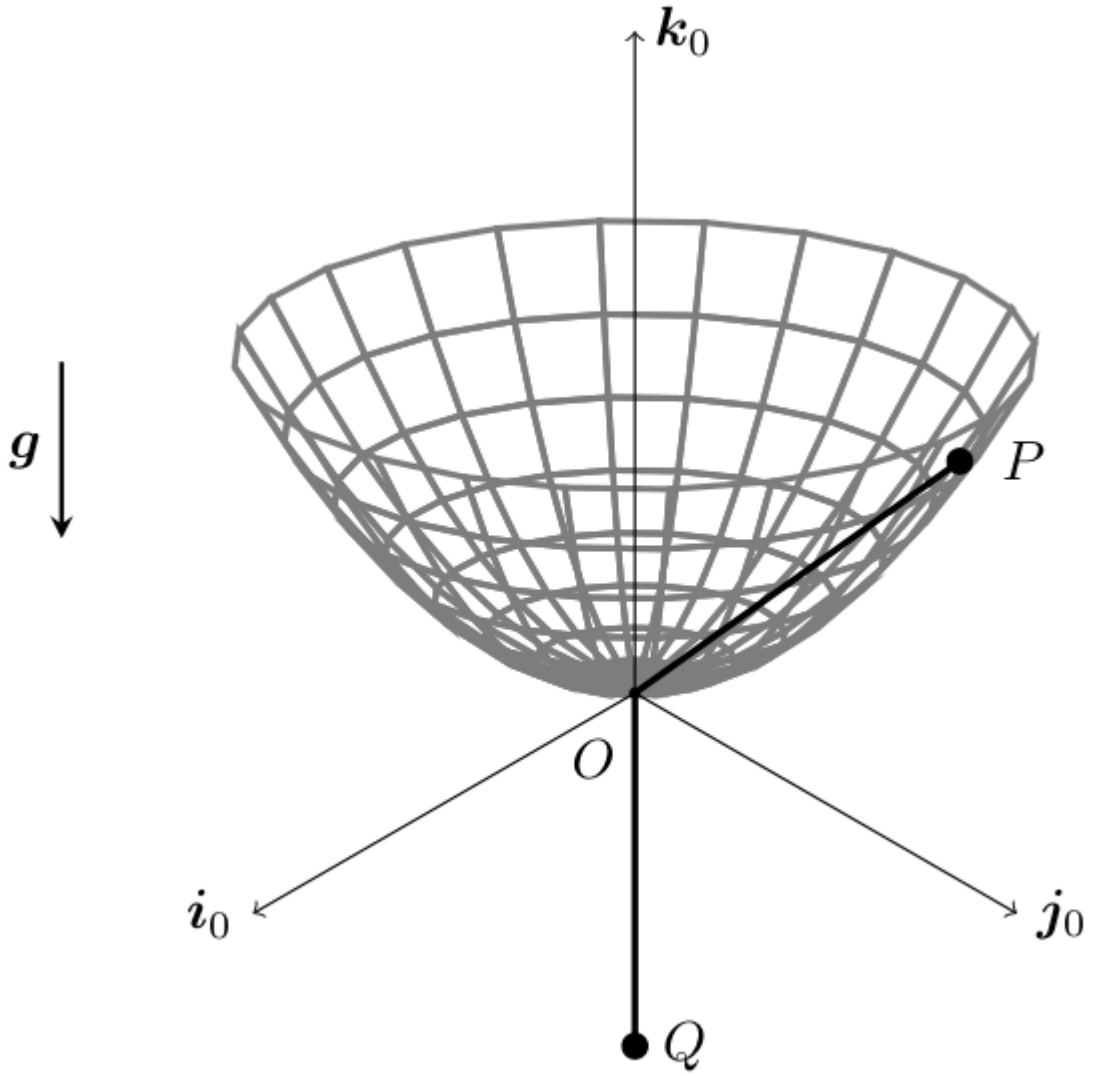


Figure: Paraboloid with P on its surface, string through the hole at O and Q hanging along the vertical axis k0 (Problems.pdf, p. 33).

(a) How many effective degrees of freedom does the system have? Justify your answer.

(b) Give the expressions for the position, velocity, and acceleration of P and Q with respect to S_0 as a function of R, θ and their derivatives. If you use any vector basis other than $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ to write your solution, define it first.

(c) Find a normal vector and two linearly independent tangent vectors to the paraboloid as a function of R, θ. They do not need to be unitary.

(d) Identify and write down the general expression of all the forces acting on P and Q, introducing new unknowns as needed.

(e) Find the condition that the solution $R(t)$ must satisfy so that the string is indeed always in tension.

(f) Using S_0 for your analysis, write down two second-order differential equations for R and θ, that do not depend on any other unknowns.

(g) Find two scalar conservation laws (i.e., first integrals).

(h) Assuming that initially $R = 1/c, \theta = 0, \dot{R} = 0, \dot{\theta} = d$, where $d \neq 0$ is a given value, write down an equation in the form $f(R) = 0$ whose roots determine the minimum and maximum values of R in the subsequent motion. You do not need to solve this equation.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two particles P and Q of mass m connected by an inextensible massless cord of length ℓ.
- Particle P is constrained to the frictionless paraboloid $z = cR^2$ ($c > 0$).
- Particle Q hangs along the vertical axis Oz.
- Cord passes through a frictionless hole at origin O.
- **CDOFs:** 2 degrees of freedom: (R, θ).
- **Initial State:** $R(0) = 1/c, \theta(0) = 0, \dot{R}(0) = 0, \dot{\theta}(0) = d \neq 0$.
- **Parameters:** m, c, ℓ, g, d.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Cylindrical basis: $\{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{k}\}$.
- Position of P: $\mathbf{r}_P = R\mathbf{e}_R + cR^2\mathbf{k}$.
- Tangent vectors to paraboloid:

$$\mathbf{t}_R = \frac{\partial \mathbf{r}_P}{\partial R} = \mathbf{e}_R + 2cR\,\mathbf{k}$$

$$\mathbf{t}_\theta = \frac{\partial \mathbf{r}_P}{\partial \theta} = R\mathbf{e}_\theta$$

- Normal vector to paraboloid:

$$\mathbf{n} = \mathbf{t}_R \times \mathbf{t}_\theta = (\mathbf{e}_R + 2cR\,\mathbf{k}) \times (R\mathbf{e}_\theta) = R\mathbf{k} - 2cR^2\,\mathbf{e}_R = R(-2cR\,\mathbf{e}_R + \mathbf{k})$$

- Position of Q: Distance along cord from O to P along generator:

Cord length constraint: $s_P + s_Q = \ell$.

For small hole at origin with straight cord to P: $|\mathbf{r}_P| = \sqrt{R^2 + c^2R^4} = R\sqrt{1 + c^2R^2}$.

Position of Q: $\mathbf{r}_Q = -(\ell - R\sqrt{1 + c^2R^2})\mathbf{k}$.

- **Change of basis (cylindrical basis).** $\mathcal{B}_1 = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ rotated by $\theta(t)$ about $\mathbf{k}$ ($\mathbf{k} = \mathbf{k}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Surface-adapted basis.** With $s = \sqrt{1 + 4c^2R^2}$, the unit meridian tangent is $\mathbf{e}_g = (\mathbf{e}_R + 2cR\,\mathbf{k})/s$ (proportional to $\mathbf{t}_R$), the unit normal is $\hat{\mathbf{n}} = (-2cR\,\mathbf{e}_R + \mathbf{k})/s$ (proportional to $\mathbf{n}$ above) and $\mathbf{e}_g \times \mathbf{e}_\theta = \hat{\mathbf{n}}$, so $\{\mathbf{e}_g, \mathbf{e}_\theta, \hat{\mathbf{n}}\}$ is right-handed and

$$[{}_1R_2] = \begin{pmatrix} 1/s & 0 & -2cR/s \\ 0 & 1 & 0 \\ 2cR/s & 0 & 1/s \end{pmatrix}, \qquad \det[{}_1R_2] = \frac{1+4c^2R^2}{s^2} = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

- Both determinants and products RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

Part (a)-(d): Forces and Vectors

1. Forces on P: Gravity $-mg\mathbf{k}$, normal reaction $\mathbf{N} = N\frac{-2cR\mathbf{e}_R+\mathbf{k}}{\sqrt{1+4c^2R^2}}$, cord tension $\mathbf{T}_P = -T\frac{\mathbf{r}_P}{|\mathbf{r}_P|}$.

2. Forces on Q: Gravity $-mg\mathbf{k}$, cord tension $\mathbf{T}_Q = +T\mathbf{k}$.

Part (e): Tension Condition

The string remains in tension as long as $T(t) > 0$.

For Q, $m\ddot{z}_Q = T - mg \implies T = m(g + \ddot{z}_Q) > 0 \iff \ddot{z}_Q > -g$.

Part (f)-(g): Two First Integrals of Motion

1. **Azimuthal First Integral (cyclic coordinate θ):**

In cylindrical coordinates $a_\theta = \frac{1}{R}\frac{d}{dt}(R^2\dot{\theta})$. The paraboloid is axisymmetric, so its reaction has no $\mathbf{e}_\theta$ component; gravity is vertical and the tension of the cord (directed from P to the hole O on the axis) lies in the meridian plane of P. The $\mathbf{e}_\theta$ equation is $\frac{m}{R}\frac{d}{dt}(R^2\dot{\theta}) = 0$:

$$C_z = mR^2\dot{\theta} = \text{constant} = m[R(0)]^2 d = m\frac{d}{c^2}$$

$$\dot{\theta}(t) = \frac{d}{c^2R^2}$$

2. **Total Mechanical Energy Conservation:**

Constraints are frictionless and scleronomic:

$$E = T_P + T_Q + V_{g,P} + V_{g,Q} = \text{constant}$$

$$T_P = \frac{1}{2}m\left[\dot{R}^2(1 + 4c^2R^2) + R^2\dot{\theta}^2\right]$$

$$T_Q = \frac{1}{2}m\dot{z}_Q^2$$

$$V = mg(cR^2) + mgz_Q$$

Part (h): Turning Point Equation $f(R) = 0$

Setting $\dot{R} = 0$ in the conserved energy equation $E(R, \dot{R} = 0) = E_0$ yields the implicit algebraic equation $f(R) = 0$ whose positive roots define the minimum and maximum radii R_{min} and R_{max}.

PHASE 4

Physical Interpretation & Dimensional Verification

- The particle executes bounded precessing orbits on the paraboloid, bounded between two parallel circles R_{min} and R_{max}.
- **Dimensional check:** $[c] = \mathrm{m}^{-1}$, $[d] = \mathrm{s}^{-1}$, $R(0) = 1/c$ is a length; $[cR^2] = \mathrm{m}$ and $[c\,R] = 1$, so $\sqrt{1 + c^2R^2}$ is dimensionless.
- **Explicit turning-point equation (part (h)).** At $\dot{R} = 0$ the kinetic energy reduces to $\frac{1}{2}mR^2\dot{\theta}^2$ with $R^2\dot{\theta}^2 = d^2/(c^4R^2)$ (from the azimuthal first integral), and the potential energy is $mg\left[cR^2 + R\sqrt{1 + c^2R^2}\right] - mg\ell$ (the term $-mg\ell$ cancels in $E - E_0$). Equating with the initial energy at $R(0) = 1/c$ (where $R\sqrt{1 + c^2R^2} = \sqrt{2}/c$), per unit mass:

$$f(R) = \frac{d^2}{2c^4R^2} + g\left(cR^2 + R\sqrt{1+c^2R^2}\right) - \frac{d^2}{2c^2} - \frac{(1+\sqrt{2})\,g}{c} = 0$$

Each term has units $\mathrm{m}^2/\mathrm{s}^2$. The allowed radii are those with $f(R) \le 0$, and $R = 1/c$ is a root (sympy: $f(1/c) = 0$).

- **Numerical example ($c = d = g = 1$):** the roots of f are $R = 0.4828$ and $R = 1$ (the launch point is the outer turning point), $f(0.7) = -0.549$ and $f(0.9) = -0.276$ are negative (allowed band between the roots), and $f(0.4) = 0.802$, $f(1.2) = 0.747$ are positive (forbidden), so the orbit is bounded between $R_{min} = 0.4828/c$ and $R_{max} = 1/c$.

Source: Problems.pdf (Page 33)